ENGINEERING ECONOMICS (CE 655)

Chapter 5: Replacement Analysis

Questions & Answers — TU Past Exam Papers (2070–2082)

CHAPTER 5 SYLLABUS OVERVIEW (6 Hours)

a) Introduction — Reasons for Replacement, Defender vs. Challenger
b) Sunk Cost Principle in Replacement
c) Economic Service Life (ESL) — Marginal Cost Analysis
d) Replacement Analysis: Defender vs. Challenger (equal & unequal lives)
e) Replacement Analysis Under a Specified Study Period
f) One-Year Replacement Decision Rule
g) Opportunity Cost Approach

SECTION A: Fundamental Concepts — Replacement Analysis

Q1. Define replacement analysis. What are the various reasons an asset may need to be replaced? Explain the terms 'defender' and 'challenger'. [2081 Ashwin Q5a, 2079 Jestha Q5a, 2075 Bhadra Q5b — 4 Marks]

Answer:

Replacement Analysis — Definition

Replacement analysis is the systematic process of determining the optimal time to retire (replace or abandon) an existing asset and substitute it with a new, better, or different asset. It answers the question: 'Should the existing asset be kept for another period, or should it be replaced now with a new alternative?'

Replacement analysis is one of the most important and common applications of engineering economics because assets inevitably deteriorate, become obsolete, or become uneconomical over time.

Reasons for Replacement (Why Assets Need to be Replaced):

1. Physical Deterioration / Wear and Tear: Assets degrade with use and age — components wear out, efficiency drops, and maintenance costs rise. Eventually the cost of keeping the asset exceeds the cost of replacement.
2. Obsolescence: Technological advancement produces newer, more efficient, more capable assets. The existing asset may still function physically but is economically inferior to available alternatives (e.g., replacing manual drafting tools with CAD software).
3. Inadequacy / Insufficient Capacity: The existing asset can no longer handle the required workload due to growth in demand or changes in requirements. A small pump that was adequate 10 years ago may be insufficient for today's demand.
4. Excessive Operating and Maintenance Costs: As assets age, operating costs (fuel, labour, repairs) and maintenance costs increase significantly. When these costs exceed the cost of owning a new asset, replacement is justified.
5. Loss of Safety and Reliability: Aging assets may become unreliable, causing frequent breakdowns that disrupt production or endanger workers. Safety regulations may mandate replacement.

6. Legal and Environmental Requirements: Changes in regulations may require replacement of assets that no longer comply with emission standards, safety codes, or environmental laws.
7. Changes in Product or Service Requirements: The nature of the product or service may change, requiring different equipment or technology that the existing asset cannot provide.

Defender and Challenger — Definitions

In replacement analysis, there are always exactly two types of alternatives being compared:

Term	Definition	Key Characteristics
Defender	The existing (currently installed) asset that is a candidate for replacement.	• Already owned • Has a current market value (opportunity cost) • Sunk costs are irrelevant • Operating costs typically increasing with age
Challenger	The best available new alternative that could replace the defender.	• Being newly purchased • First cost is the actual purchase price • Represents the 'best' new option currently available • Operating costs typically lower initially

Key Principle: The decision is always: 'Is it more economical to keep the defender for one more period (or until its Economic Service Life), or to replace it now with the challenger?'

In the opportunity cost approach, the defender's current market value (trade-in or resale value) is treated as its 'first cost' — because retaining the defender means forgoing the cash that could be received by selling it today.

Q2. Explain sunk cost. Why should sunk cost be ignored in replacement analysis? Illustrate with an example. [2082 Kartik Q5b, 2080 Chaitra Q5a, 2078 Chaitra Q5a, 2074 Bhadra Q5b — 4 Marks]

Answer:

Sunk Cost — Definition

A sunk cost is a cost that has already been incurred in the past and cannot be recovered regardless of any future decision. It is a past, irreversible expenditure that has no bearing on future cash flows.

Example: A company purchased a machine 3 years ago for Rs. 1,20,000. Its current market value is Rs. 30,000. The original purchase price of Rs. 1,20,000, and the depreciation already taken, are sunk costs. The company cannot recover that Rs. 1,20,000 regardless of whether it keeps or replaces the machine.

Why Sunk Costs Must Be Ignored in Replacement Analysis

Sunk costs must be excluded from replacement analysis because:

1. They do not affect future cash flows. The amount already spent on the defender cannot be changed by any current decision. Including it would distort the economic comparison.
2. Rational decision-making requires comparing only future costs and benefits. The only relevant costs are those that will differ between keeping and replacing.
3. Including sunk costs creates the 'sunk cost fallacy' — the irrational tendency to continue with a losing investment just because 'we've already spent so much on it.' This leads to economically poor decisions.
4. The correct 'first cost' of the defender is its current market value (opportunity cost) — not its book value or original purchase price.

Illustrative Example: The Sunk Cost Fallacy

A construction company purchased an excavator 4 years ago for Rs. 5,00,000 (original cost). Its current book value (after 4 years of depreciation) is Rs. 2,00,000. Its actual current market resale value is Rs. 80,000.

A new challenger excavator costs Rs. 3,50,000 with lower operating costs.

Incorrect approach (including sunk cost): 'We paid Rs. 5,00,000 for it. Its book value is Rs. 2,00,000. We can't replace it — we haven't recovered our investment.'

Correct approach (opportunity cost): The defender's relevant 'first cost' for future analysis is its current market value = Rs. 80,000. The Rs. 5,00,000 original cost and Rs. 2,00,000 book value are sunk — they should be completely ignored. The decision must be based on comparing future AW of the defender (using Rs. 80,000 as its starting value) with the AW of the challenger (Rs. 3,50,000 first cost).

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Defender's first cost for analysis = Current Market Value = Rs. 80,000
(NOT Rs. 5,00,000 or Rs. 2,00,000)
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Rule: In replacement analysis, ALWAYS use current market value as the defender's first cost. Original cost and book value are sunk costs and must be ignored.

Q3. Explain the concept of Economic Service Life (ESL). What is marginal cost? How is ESL determined using the marginal cost approach? [2081 Chaitra Q5a, 2079 Ashwin Q5a, 2077 Chaitra Q5a — 4 Marks]

Answer:

Economic Service Life (ESL) — Definition

The Economic Service Life (ESL) — also called the Optimal Replacement Interval or Economic Life — is the number of years at which the Total Annual Worth (or Annual Equivalent Cost) of owning and operating an asset is minimised. It is the period for which the asset should be kept to achieve the lowest equivalent annual cost.

Every asset has two conflicting cost trends:

- Capital Recovery (CR) Cost: Decreases over time — the longer an asset is kept, the lower the annual capital recovery cost (the initial investment is spread over more years).
- Annual Operating & Maintenance (O&M) Cost: Increases over time — as the asset ages, it requires more repairs, consumes more fuel, and becomes less efficient.

ESL is the point where Total Annual Cost (CR + O&M) is at its minimum — where the declining CR cost and rising O&M cost intersect at their lowest combined total.

Marginal Cost (Year-by-Year Cost)

Marginal Cost (also called the cost of keeping an asset for one more year) is the total cost incurred if an asset is kept for exactly one additional year. It consists of:

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Marginal Cost (year k) = Loss in Market Value + Interest on Opening Market
Value + O&M Cost for year k
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$$MC_k = (MV_{k-1} - MV_k) + MV_{k-1} \times i + O\&M_k$$

Where: MV_{k-1} = market value at beginning of year k (= end of year k-1), MV_k = market value at end of year k, i = MARR

Alternatively, the marginal cost can be expressed as the AW of keeping the asset for exactly k years from new:

$$TC(k) = [P - MV_k(P/F,i,k)](A/P,i,k) + MV_k \times i + \text{AW of O\&M costs over k years}$$

Determining ESL Using Marginal Cost Analysis:

1. Obtain market values (MV) and O&M costs for each year of the asset's potential life.
2. Compute the Total Annual Cost (TC) for keeping the asset exactly n years, for n = 1, 2, 3, ... years.
3. TC(n) = Capital Recovery Cost(n) + AW of O&M Costs(n)
4. CR(n) = (P - MV_n)(A/P,i,n) + MV_n × i
5. AW of O&M = [O&M_1(P/F,i,1) + O&M_2(P/F,i,2) + ... + O&M_n(P/F,i,n)](A/P,i,n)
6. The year n* that gives the minimum TC(n) is the ESL.

Alternative (Marginal Cost Approach for ESL): The ESL is the year just before the marginal cost for the next year exceeds the minimum TC achieved so far. Keep the asset as long as the marginal cost for next year is less than the total average cost for the current best year.

Year (k)	Market Value (MV)	O&M Cost	Capital Recovery (CR)	AW of O&M	Total AW (TC)
1	↓ Decreasing	↑ Increasing	High (investment not spread)	Low	Often high
ESL (n*)	Minimum TC	—	Declining	Rising	MINIMUM ← ESL
Beyond ESL	Very low	Very high	Very low	Dominant	Rising again

SECTION B: Economic Service Life — Numerical Problems

Q4. Determine the economic service life of the following asset. First Cost = Rs. 10,000; O&M costs and market values are as follows. MARR = 10%. [2082 Kartik Q5a, 2080 Ashwin Q5b — 6 Marks]

Answer:

Year (k)	Market Value at End of Year (Rs.)	O&M Cost in Year k (Rs.)
1	7,000	2,000
2	5,000	3,000
3	3,500	4,500
4	2,500	6,000
5	1,500	8,000

P = Rs. 10,000, MARR = i = 10%

Total Annual Cost (TC) for each retention period n:

Formula: TC(n) = [P - MV_n](A/P,i,n) + MV_n × i + AW(O&M over n years)

(A/P factors: n=1→1.1000, n=2→0.57619, n=3→0.40211, n=4→0.31547, n=5→0.26380)

PW(O&M) factors: (P/F,10%,1)=0.9091, (P/F,10%,2)=0.8264, (P/F,10%,3)=0.7513, (P/F,10%,4)=0.6830, (P/F,10%,5)=0.6209)

n = 1 year:

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CR(1) = (10,000-7,000)(1.1000) + 7,000(0.10) = 3,000(1.10) + 700 = 3,300 +
                                 700 = Rs. 4,000

          AW_OM(1) = 2,000     [only one year, AW = actual cost]

                    TC(1) = 4,000 + 2,000 = Rs. 6,000
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n = 2 years:

CR(2) = (10,000-5,000)(0.57619) + 5,000(0.10) = 5,000(0.57619) + 500 = 2,881 + 500 = Rs. 3,381

PW_OM(2) = 2,000(0.9091)+3,000(0.8264) = 1,818+2,479 = 4,297

AW_OM(2) = 4,297 × 0.57619 = Rs. 2,476

TC(2) = 3,381 + 2,476 = Rs. 5,857

n = 3 years:

CR(3) = (10,000-3,500)(0.40211) + 3,500(0.10) = 6,500(0.40211) + 350 = 2,614 + 350 = Rs. 2,964

PW_OM(3) = 2,000(0.9091)+3,000(0.8264)+4,500(0.7513) = 1,818+2,479+3,381 = 7,678

AW_OM(3) = 7,678 × 0.40211 = Rs. 3,088

TC(3) = 2,964 + 3,088 = Rs. 6,052

n = 4 years:

CR(4) = (10,000-2,500)(0.31547) + 2,500(0.10) = 7,500(0.31547) + 250 = 2,366 + 250 = Rs. 2,616

PW_OM(4) = 7,678 + 6,000(0.6830) = 7,678 + 4,098 = 11,776

AW_OM(4) = 11,776 × 0.31547 = Rs. 3,715

TC(4) = 2,616 + 3,715 = Rs. 6,331

n = 5 years:

CR(5) = (10,000-1,500)(0.26380) + 1,500(0.10) = 8,500(0.26380) + 150 = 2,242 + 150 = Rs. 2,392

PW_OM(5) = 11,776 + 8,000(0.6209) = 11,776 + 4,967 = 16,743

AW_OM(5) = 16,743 × 0.26380 = Rs. 4,416

TC(5) = 2,392 + 4,416 = Rs. 6,808

Retention (n years)	CR Cost (Rs.)	AW of O&M (Rs.)	Total AW / TC (Rs.)
1	4,000	2,000	6,000
2	3,381	2,476	5,857 ← Minimum
3	2,964	3,088	6,052
4	2,616	3,715	6,331
5	2,392	4,416	6,808

Economic Service Life (ESL) = 2 years (minimum TC = Rs. 5,857 at n = 2).

The asset should be replaced every 2 years to minimise the equivalent annual cost of ownership.

Q5. Find the economic life and minimum annual cost of the following machine. First Cost = Rs. 20,000; Salvage decreases Rs. 3,000/year; O&M = Rs. 2,000 in year 1 increasing Rs. 1,500/year. MARR = 12%. [2081 Chaitra Q5b, 2079 Ashwin Q5b — 6 Marks]

Answer:

P = Rs. 20,000. MV decreases by Rs. 3,000/year: MV(1)=17,000, MV(2)=14,000, MV(3)=11,000, MV(4)=8,000, MV(5)=5,000, MV(6)=2,000

O&M: A1=2,000, G=1,500/year: O&M(1)=2,000, O&M(2)=3,500, O&M(3)=5,000, O&M(4)=6,500, O&M(5)=8,000, O&M(6)=9,500

MARR = 12%

(A/P factors: 1→1.12, 2→0.59170, 3→0.41635, 4→0.32923, 5→0.27741, 6→0.24323)

(P/F factors: 1→0.8929, 2→0.7972, 3→0.7118, 4→0.6355, 5→0.5674, 6→0.5066)

TC(n) = CR(n) + AW of O&M(n)

n = 1:

CR(1) = (20,000-17,000)(1.12)+17,000(0.12) = 3,360+2,040 = Rs. 5,400

AW_OM(1) = 2,000

TC(1) = Rs. 7,400

n = 2:

CR(2) = (20,000-14,000)(0.59170)+14,000(0.12) = 3,550+1,680 = Rs. 5,230

PW_OM(2) = 2,000(0.8929)+3,500(0.7972) = 1,786+2,790 = 4,576

AW_OM(2) = 4,576×0.59170 = Rs. 2,707

TC(2) = 5,230+2,707 = Rs. 7,937

n = 3:

CR(3) = (20,000-11,000)(0.41635)+11,000(0.12) = 3,747+1,320 = Rs. 5,067

PW_OM(3) = 4,576+5,000(0.7118) = 4,576+3,559 = 8,135

AW_OM(3) = 8,135×0.41635 = Rs. 3,388

TC(3) = 5,067+3,388 = Rs. 8,455

n = 4:

CR(4) = (20,000-8,000)(0.32923)+8,000(0.12) = 3,951+960 = Rs. 4,911

PW_OM(4) = 8,135+6,500(0.6355) = 8,135+4,131 = 12,266

AW_OM(4) = 12,266×0.32923 = Rs. 4,039

TC(4) = 4,911+4,039 = Rs. 8,950

n = 5:

CR(5) = (20,000-5,000)(0.27741)+5,000(0.12) = 4,161+600 = Rs. 4,761

PW_OM(5) = 12,266+8,000(0.5674) = 12,266+4,539 = 16,805

AW_OM(5) = 16,805×0.27741 = Rs. 4,662

TC(5) = 4,761+4,662 = Rs. 9,423

n = 6:

CR(6) = (20,000-2,000)(0.24323)+2,000(0.12) = 4,378+240 = Rs. 4,618

PW_OM(6) = 16,805+9,500(0.5066) = 16,805+4,813 = 21,618

AW_OM(6) = 21,618×0.24323 = Rs. 5,258

TC(6) = 4,618+5,258 = Rs. 9,876

n (years)	CR (Rs.)	AW of O&M (Rs.)	Total TC (Rs.)

1	5,400	2,000	7,400 ← Minimum
2	5,230	2,707	7,937
3	5,067	3,388	8,455
4	4,911	4,039	8,950
5	4,761	4,662	9,423
6	4,618	5,258	9,876

Economic Service Life (ESL) = 1 year (minimum TC = Rs. 7,400 at n = 1).

This result indicates the machine is most economical when replaced every year. The high initial O&M cost and rapidly declining market value combine to make longer retention increasingly expensive relative to the capital recovery savings. The challenger should be considered for immediate replacement.

Q6. A machine has a first cost of Rs. 50,000. The market value and O&M costs are given below. Find the ESL at MARR = 10%. [2079 Chaitra Q5a, 2077 Chaitra Q5b — 6 Marks]

Answer:

Year	Market Value (Rs.)	Annual O&M (Rs.)
0 (Purchase)	50,000	—
1	40,000	6,000
2	32,000	8,000
3	25,000	11,000
4	18,000	14,000
5	12,000	18,000
6	8,000	22,000

P = Rs. 50,000, MARR = 10%

(A/P,10%,n): n1=1.1000, n2=0.57619, n3=0.40211, n4=0.31547, n5=0.26380, n6=0.22961

(P/F,10%,k): k1=0.9091, k2=0.8264, k3=0.7513, k4=0.6830, k5=0.6209, k6=0.5645

Compute TC(n) = CR(n) + AW_OM(n):

n=1:

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CR(1)=(50,000-40,000)(1.10)+40,000(0.10) = 11,000+4,000 = 15,000
AW_OM(1)=6,000    |    TC(1) = Rs. 21,000
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n=2:

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CR(2)=(50,000-32,000)(0.57619)+32,000(0.10) = 10,371+3,200 = 13,571
PW_OM = 6,000(0.9091)+8,000(0.8264) = 5,455+6,611 = 12,066
AW_OM(2) = 12,066×0.57619 = 6,952    |    TC(2) = Rs. 20,523
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n=3:

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CR(3)=(50,000-25,000)(0.40211)+25,000(0.10) = 10,053+2,500 = 12,553
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PW_OM = 12,066+11,000(0.7513) = 12,066+8,264 = 20,330

AW_OM(3) = 20,330×0.40211 = 8,175 | TC(3) = Rs. 20,728

n=4:

CR(4)=(50,000-18,000)(0.31547)+18,000(0.10) = 10,095+1,800 = 11,895

PW_OM = 20,330+14,000(0.6830) = 20,330+9,562 = 29,892

AW_OM(4) = 29,892×0.31547 = 9,431 | TC(4) = Rs. 21,326

n=5:

CR(5)=(50,000-12,000)(0.26380)+12,000(0.10) = 10,024+1,200 = 11,224

PW_OM = 29,892+18,000(0.6209) = 29,892+11,176 = 41,068

AW_OM(5) = 41,068×0.26380 = 10,834 | TC(5) = Rs. 22,058

n=6:

CR(6)=(50,000-8,000)(0.22961)+8,000(0.10) = 9,644+800 = 10,444

PW_OM = 41,068+22,000(0.5645) = 41,068+12,419 = 53,487

AW_OM(6) = 53,487×0.22961 = 12,281 | TC(6) = Rs. 22,725

n	CR (Rs.)	AW_OM (Rs.)	TC (Rs.)
1	15,000	6,000	21,000
2	13,571	6,952	20,523 ← Minimum
3	12,553	8,175	20,728
4	11,895	9,431	21,326
5	11,224	10,834	22,058
6	10,444	12,281	22,725

ESL = 2 years, Minimum Annual Cost = Rs. 20,523.

Q7. A new machine costs Rs. 30,000 and has the following operating costs and salvage values. Determine ESL at MARR = 15%. [2076 Baisakh Q5a — 5 Marks]

Answer:

Year	End-of-Year MV (Rs.)	Annual O&M (Rs.)
1	22,000	4,000
2	16,000	5,500
3	11,000	7,500
4	7,000	10,000
5	4,000	13,000

P = Rs. 30,000, MARR = 15%

(A/P,15%,n): 1→1.15, 2→0.61512, 3→0.43798, 4→0.35027, 5→0.29832

Summary of TC(n):

n=1: CR=(30,000-22,000)(1.15)+22,000(0.15)=9,200+3,300=12,500; AW_OM=4,000; TC=Rs. 16,500

n=2:

CR=(30,000-16,000)(0.61512)+16,000(0.15)=8,611+2,400=11,011

PW_OM=4,000(0.8696)+5,500(0.7561)=3,478+4,159=7,637;
AW_OM=7,637×0.61512=4,697

TC(2) = Rs. 15,708 ← Minimum so far

n=3:

CR=(30,000-11,000)(0.43798)+11,000(0.15)=8,322+1,650=9,972

PW_OM=7,637+7,500(0.6575)=7,637+4,931=12,568; AW_OM=12,568×0.43798=5,505

TC(3) = Rs. 15,477 ← Minimum so far

n=4:

CR=(30,000-7,000)(0.35027)+7,000(0.15)=8,056+1,050=9,106

PW_OM=12,568+10,000(0.5718)=12,568+5,718=18,286;
AW_OM=18,286×0.35027=6,409

TC(4) = Rs. 15,515

n=5:

CR=(30,000-4,000)(0.29832)+4,000(0.15)=7,756+600=8,356

PW_OM=18,286+13,000(0.4972)=18,286+6,464=24,750;
AW_OM=24,750×0.29832=7,384

TC(5) = Rs. 15,740

n	CR (Rs.)	AW_OM (Rs.)	TC (Rs.)
1	12,500	4,000	16,500
2	11,011	4,697	15,708
3	9,972	5,505	15,477 ← Minimum
4	9,106	6,409	15,515
5	8,356	7,384	15,740

ESL = 3 years, Minimum Annual Cost = Rs. 15,477.

SECTION C: Replacement Decision — Defender vs. Challenger (AW Method)

Q8. A 3-year old machine (defender) has a current market value of Rs. 40,000. Its expected O&M costs and future market values are as follows. A new challenger machine costs Rs. 90,000 with O&M costs and market values as given. Should the defender be replaced now? MARR = 10%. [2082 Kartik Q6, 2080 Ashwin Q6 — 8 Marks]

Answer:

Year from Now	Defender MV (Rs.)	Defender O&M (Rs.)	Challenger MV (Rs.)	Challenger O&M (Rs.)
0 (now)	40,000	—	90,000	—
1	28,000	12,000	65,000	8,000
2	18,000	15,000	46,000	10,000
3	10,000	20,000	32,000	13,000
4	5,000	26,000	22,000	17,000
5	—	—	15,000	22,000

MARR = 10%

Step 1: Find ESL and Minimum AW Cost of Defender

Defender first cost = current market value = Rs. 40,000 (opportunity cost approach)

(A/P,10%,n): 1→1.10, 2→0.57619, 3→0.40211, 4→0.31547

Defender TC(n):

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TC_D(1): CR=(40,000-28,000)(1.10)+28,000(0.10)=13,200+2,800=16,000;
OM=12,000; TC=Rs. 28,000
TC_D(2): CR=(40,000-18,000)(0.57619)+18,000(0.10)=12,676+1,800=14,476
PW_OM=12,000(0.9091)+15,000(0.8264)=10,909+12,396=23,305
AW_OM=23,305×0.57619=13,429; TC_D(2)=14,476+13,429=Rs. 27,905 ← Min
Defender
TC_D(3): CR=(40,000-10,000)(0.40211)+10,000(0.10)=12,063+1,000=13,063
PW_OM=23,305+20,000(0.7513)=23,305+15,026=38,331
AW_OM=38,331×0.40211=15,413; TC_D(3)=13,063+15,413=Rs. 28,476
TC_D(4): CR=(40,000-5,000)(0.31547)+5,000(0.10)=11,041+500=11,541
PW_OM=38,331+26,000(0.6830)=38,331+17,758=56,089
AW_OM=56,089×0.31547=17,697; TC_D(4)=11,541+17,697=Rs. 29,238
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Defender ESL = 2 years, Minimum AW Cost of Defender = Rs. 27,905

Step 2: Find ESL and Minimum AW Cost of Challenger

Challenger first cost P = Rs. 90,000

(A/P,10%,n): 3→0.40211, 4→0.31547, 5→0.26380

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TC_C(1): CR=(90,000-65,000)(1.10)+65,000(0.10)=27,500+6,500=34,000;
OM=8,000; TC=Rs. 42,000
TC_C(2): CR=(90,000-46,000)(0.57619)+46,000(0.10)=25,352+4,600=29,952
PW_OM=8,000(0.9091)+10,000(0.8264)=7,273+8,264=15,537
AW_OM=15,537×0.57619=8,952; TC_C(2)=29,952+8,952=Rs. 38,904
TC_C(3): CR=(90,000-32,000)(0.40211)+32,000(0.10)=23,322+3,200=26,522
PW_OM=15,537+13,000(0.7513)=15,537+9,767=25,304
AW_OM=25,304×0.40211=10,177; TC_C(3)=26,522+10,177=Rs. 36,699
TC_C(4): CR=(90,000-22,000)(0.31547)+22,000(0.10)=21,452+2,200=23,652
PW_OM=25,304+17,000(0.6830)=25,304+11,611=36,915
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AW_OM=36,915×0.31547=11,648; TC_C(4)=23,652+11,648=Rs. 35,300

TC_C(5): CR=(90,000-15,000)(0.26380)+15,000(0.10)=19,785+1,500=21,285

PW_OM=36,915+22,000(0.6209)=36,915+13,660=50,575

AW_OM=50,575×0.26380=13,342; TC_C(5)=21,285+13,342=Rs. 34,627

TC_C(5) is lower than TC_C(4). Check n=6 (if data available) or accept n=5 as minimum based on trend.

Challenger ESL ≈ 5 years (continuing trend), Minimum AW Cost of Challenger ≈ Rs. 34,627

Step 3: Replacement Decision

Compare Minimum AW of Defender vs. Minimum AW of Challenger:

Min AW_Defender = Rs. 27,905 vs. Min AW_Challenger = Rs. 34,627

Min AW of Defender (Rs. 27,905) < Min AW of Challenger (Rs. 34,627).

Decision: DO NOT REPLACE the defender now. Keep the defender for its ESL of 2 more years before reconsidering replacement.

The challenger is more expensive to own annually than the defender at their respective optimal lives. Replacement is not economically justified at this time.

Q9. A currently owned machine (defender) was bought 2 years ago for Rs. 80,000. Its current market value is Rs. 35,000. Annual O&M for the defender will be Rs. 18,000/year. A new machine (challenger) costs Rs. 60,000, has O&M of Rs. 10,000/year, and a salvage value of Rs. 8,000 after 8 years. Both have the same remaining / total life. MARR = 12%. Should the machine be replaced? [2081 Ashwin Q6a, 2079 Jestha Q6 — 6 Marks]

Answer:

Defender: Current MV = Rs. 35,000 (first cost for analysis — sunk cost principle); Remaining life = 6 years; MV at end of 6 years = Rs. 5,000 (assumed); Annual O&M = Rs. 18,000

Challenger: P = Rs. 60,000; Life = 8 years; S = Rs. 8,000; Annual O&M = Rs. 10,000; MARR = 12%

Note: For comparison with equal remaining life of 6 years, assume challenger is used for 6 years with market value Rs. 12,000 at year 6 (interpolated).

AW of Defender (6 years, MARR = 12%):

(A/P,12%,6) = 0.24323, (A/F,12%,6) = 0.12323

AW_D = -35,000(A/P,12%,6) + 5,000(A/F,12%,6) - 18,000

= -35,000(0.24323) + 5,000(0.12323) - 18,000

= -8,513 + 616 - 18,000 = -Rs. 25,897

AW of Challenger (using 6-year study period, S=12,000 at year 6):

AW_C = -60,000(0.24323) + 12,000(0.12323) - 10,000

= -14,594 + 1,479 - 10,000 = -Rs. 23,115

AW_C (-Rs. 23,115) < AW_D (-Rs. 25,897) in absolute cost terms (challenger is cheaper annually).

Decision: REPLACE the defender with the challenger now. The challenger saves Rs. 25,897 - Rs. 23,115 = Rs. 2,782 per year compared to keeping the defender.

Note: The original purchase price of Rs. 80,000 (paid 2 years ago) and the book value are irrelevant sunk costs. Only the current market value of Rs. 35,000 is used as the defender's first cost.

Q10. An existing pump (defender) has a current market value of Rs. 15,000, O&M costs of Rs. 25,000/year, and no significant salvage value, with 5 more years of use. A new pump (challenger) costs Rs. 65,000, has O&M costs of Rs. 12,000/year, a salvage value of Rs. 10,000, and a life of 10 years. Should the existing pump be replaced? MARR = 10%. [2079 Chaitra Q5b, 2077 Chaitra Q6 — 6 Marks]

Answer:

Defender: MV_now = Rs. 15,000, O&M = Rs. 25,000/year, remaining life = 5 years, S = 0

Challenger: P = Rs. 65,000, O&M = Rs. 12,000/year, life = 10 years, S = Rs. 10,000

MARR = 10%. Unequal lives — use AW method (repeatability assumption).

$$(A/P,10\%,5)=0.26380;\ (A/F,10\%,5)=0.16380$$

$$(A/P,10\%,10)=0.16275;\ (A/F,10\%,10)=0.06275$$

AW of Defender (5 years):

$$AW_D = -15{,}000(A/P,10\%,5) + 0 - 25{,}000$$

$$= -15{,}000(0.26380) - 25{,}000 = -3{,}957 - 25{,}000 = -Rs.\ 28{,}957$$

AW of Challenger (10 years):

$$AW_C = -65{,}000(A/P,10\%,10) + 10{,}000(A/F,10\%,10) - 12{,}000$$

$$= -65{,}000(0.16275) + 10{,}000(0.06275) - 12{,}000$$

$$= -10{,}579 + 628 - 12{,}000 = -Rs.\ 21{,}951$$

AW_C (-Rs. 21,951) is cheaper than AW_D (-Rs. 28,957). The challenger saves Rs. 7,006/year.

Decision: REPLACE the existing pump with the new challenger immediately. The annual saving of Rs. 7,006 justifies the replacement.

Q11. Replacement analysis — should defender be replaced? Defender data: current MV = Rs. 20,000; O&M = Rs. 14,000/year; MV decreases Rs. 3,000/year; remaining life = 4 years. Challenger: P = Rs. 50,000; O&M = Rs. 8,000/year; S = Rs. 10,000; life = 8 years. MARR = 10%. [2075 Bhadra Q6, 2078 Baishakh Q5b — 6 Marks]

Answer:

Defender: P_D = Rs. 20,000 (current MV), O&M = Rs. 14,000, remaining life = 4 years

MV schedule: MV(1)=17,000, MV(2)=14,000, MV(3)=11,000, MV(4)=8,000

Challenger: P = Rs. 50,000, O&M = Rs. 8,000/yr, S = Rs. 10,000, n = 8 years

ESL of Defender:

(A/P,10%,n): 1→1.10, 2→0.57619, 3→0.40211, 4→0.31547

$$TC_D(1)=(20{,}000-17{,}000)(1.10)+17{,}000(0.10)+14{,}000=3{,}300+1{,}700+14{,}000=Rs.\ 19{,}000$$

$$TC_D(2):\ CR=(20{,}000-14{,}000)(0.57619)+14{,}000(0.10)=3{,}457+1{,}400=4{,}857$$

$$AW_OM=(14,000\times0.9091+14,000\times0.8264)\times0.57619=14,000(1.7355\times0.57619)=14,000\times0.9998=14,000 \text{ [uniform]}$$

$$TC_D(2)=4,857+14,000=Rs.\ 18,857 \leftarrow Min$$

$$TC_D(3):\ CR=(20,000-11,000)(0.40211)+11,000(0.10)=3,619+1,100=4,719;\ TC=18,719 \leftarrow Lower$$

$$TC_D(4):\ CR=(20,000-8,000)(0.31547)+8,000(0.10)=3,786+800=4,586;\ TC=18,586 \leftarrow Min$$

O&M is uniform Rs. 14,000/year for defender, so AW_OM = Rs. 14,000 for all n.

n	CR_D (Rs.)	AW_OM_D (Rs.)	TC_D (Rs.)
1	5,000	14,000	19,000
2	4,857	14,000	18,857
3	4,719	14,000	18,719
4	4,586	14,000	18,586 ← Minimum

Defender ESL = 4 years (full remaining life), Min TC_D = Rs. 18,586

AW of Challenger (8 years):

$$(A/P,10\%,8)=0.18744;\ (A/F,10\%,8)=0.08744$$

$$AW_C = -50,000(0.18744)+10,000(0.08744)-8,000$$

$$= -9,372+874-8,000 = -Rs.\ 16,498$$

AW_C (Rs. 16,498/yr cost) < AW_D (Rs. 18,586/yr cost). Challenger is cheaper by Rs. 2,088/year.

Decision: REPLACE the defender with the challenger now. Annual saving = Rs. 2,088/year.

SECTION D: One-Year Replacement Decision (Marginal Cost Approach)

Q12. Explain the one-year replacement decision rule. When should an asset be kept for one more year vs. replaced? [2078 Chaitra Q5b, 2076 Baisakh Q5b — 4 Marks]

Answer:

One-Year Replacement Decision Rule

The one-year replacement decision rule determines whether an existing asset (defender) should be kept for exactly one more year or replaced immediately with the challenger. The rule is based on comparing the marginal cost of keeping the defender one more year against the minimum annual cost of the challenger.

Marginal Cost of Keeping the Defender for One More Year:

$$MC_defender = (MV_0 - MV_1) + MV_0 \times i + O\&M \text{ cost for year } 1$$

Where: MV_0 = current market value of defender, MV_1 = estimated market value one year from now, i = MARR, O&M = operating and maintenance cost for the next year.

This marginal cost represents: (1) the loss in market value over the year, plus (2) the interest foregone on the current market value (opportunity cost of not selling now), plus (3) the operating cost for the year.

Decision Rule:

```
If MC_defender ≤ AW_challenger(min): Keep the defender for one more year
If MC_defender > AW_challenger(min): Replace the defender now with challenger
```

Economic Justification: The marginal cost is the cost incurred by keeping the asset for one more year. If this cost is lower than the best available challenger's annual cost, it is economical to wait. If the marginal cost exceeds the challenger's annual cost, the challenger saves money starting immediately.

Example Application:

Defender: MV_0 = Rs. 30,000, MV_1 = Rs. 20,000, O&M next year = Rs. 8,000, MARR = 10%.

```
MC = (30,000-20,000) + 30,000(0.10) + 8,000 = 10,000+3,000+8,000 = Rs. 21,000
```

Challenger minimum annual cost = Rs. 18,000.

Since MC (Rs. 21,000) > AW_challenger (Rs. 18,000), replace the defender now.

Q13. A defender has current MV = Rs. 25,000, will be worth Rs. 18,000 after one more year, and will incur O&M of Rs. 9,000 next year. The best available challenger has a minimum AW of cost Rs. 22,000 per year. MARR = 12%. What is your recommendation? [2079 Ashwin Q6a, 2074 Bhadra Q6a — 4 Marks]

Answer:

Defender: MV_0 = Rs. 25,000, MV_1 = Rs. 18,000, O&M = Rs. 9,000, MARR = 12%

Challenger: Min AW cost = Rs. 22,000/year

Marginal Cost of Defender for One More Year:

```
MC = (MV_0 - MV_1) + MV_0 × i + O&M
= (25,000 - 18,000) + 25,000(0.12) + 9,000
= 7,000 + 3,000 + 9,000 = Rs. 19,000
```

Decision:

```
MC_defender = Rs. 19,000  <  AW_challenger = Rs. 22,000
```

Since MC of defender (Rs. 19,000) < AW of challenger (Rs. 22,000), it is cheaper to keep the defender for one more year.

Recommendation: DO NOT REPLACE now. Keep the defender for at least one more year. Re-evaluate the replacement decision at the end of next year using updated market values and O&M estimates.

SECTION E: Replacement Under a Specified Study Period

Q14. An asset (defender) with 3 years of remaining life and a new challenger are being compared over a fixed 5-year study period. Determine the optimal

replacement strategy. MARR = 10%. [2081 Chaitra Q6, 2079 Chaitra Q6 — 8 Marks]

Answer:

Year	Defender MV (Rs.)	Defender O&M (Rs.)	Challenger MV (Rs.)	Challenger O&M (Rs.)
0 (now)	30,000	—	70,000	—
1	20,000	10,000	55,000	7,000
2	13,000	13,000	43,000	9,000
3	8,000	17,000	33,000	11,500
4	—	—	25,000	14,500
5	—	—	18,000	18,000

Study period = 5 years, MARR = 10%

Since the defender has only 3 years remaining, possible strategies over 5 years are:

- Strategy 1: Keep defender for 1 year (yr 1), replace with challenger for years 2-5 (4 yr challenger use)
- Strategy 2: Keep defender for 2 years (yr 1-2), replace with challenger for years 3-5 (3 yr challenger use)
- Strategy 3: Keep defender for 3 years (yr 1-3), replace with challenger for years 4-5 (2 yr challenger use)
- Strategy 4: Replace with challenger immediately — 5 years challenger use

Compute AW for each strategy over the 5-year study period (using PW then convert):

(P/F,10%,k): k=1→0.9091, k=2→0.8264, k=3→0.7513, k=4→0.6830, k=5→0.6209

(P/A,10%,5)=3.7908; (A/P,10%,5)=0.26380

Strategy 4 (Challenger from t=0, 5 years):

$$PW_4 = -70{,}000+[-7{,}000(0.9091)-9{,}000(0.8264)-11{,}500(0.7513)-14{,}500(0.6830)-18{,}000(0.6209)]+18{,}000(0.6209)$$

$$OM_PW = -6{,}364-7{,}438-8{,}640-9{,}904-11{,}176 = -43{,}522$$

$$PW_4 = -70{,}000 - 43{,}522 + 18{,}000(0.6209) = -70{,}000-43{,}522+11{,}176 = -\text{Rs. } 1{,}02{,}346$$

$$AW_4 = -1{,}02{,}346 \times 0.26380 = -\text{Rs. } 26{,}999$$

Strategy 3 (Defender 3 yr, Challenger 2 yr):

Defender years 1-3: costs/salvage based on defender data

$$PW_def = -30{,}000 - 10{,}000(0.9091)-13{,}000(0.8264)-17{,}000(0.7513)+8{,}000(0.7513)$$

$$= -30{,}000-9{,}091-10{,}743-12{,}772+6{,}010 = -56{,}596$$

Challenger purchase at t=3 (PW at t=3, then discount to t=0):

$$PW_chal@t3 = -70{,}000-14{,}500(P/F,10\%,1)-18{,}000(P/F,10\%,2)+18{,}000(P/F,10\%,2)$$

$$= -70{,}000-14{,}500(0.9091)+(-18{,}000+18{,}000)(0.8264)$$

$$= -70{,}000-13{,}182+0 = -83{,}182 \text{ [Note: salvage at end yr5 = MV at yr 2 of challenger = 43,000]}$$

Recalculate properly:

Challenger used from t=3 to t=5 (2 years). At t=3: buy for 70,000. Year 4 O&M=14,500, Year 5 O&M=18,000, MV at t=5=18,000 (yr2 of challenger life)

$$PW_chal(t3\ to\ t5)@t3 = -70{,}000-14{,}500(0.9091)-18{,}000(0.8264)+18{,}000(0.8264)...$$

Wait - challenger MV at year 2 of its life = Rs. 43,000 (not 18,000). Correcting:

$$PW_chal@t3 = -70{,}000-14{,}500(0.9091)-18{,}000(0.8264)+43{,}000(0.8264)$$

$$= -70{,}000-13{,}182-14{,}875+35{,}535 = -62{,}522$$

$$PW_chal@t0 = -62{,}522 \times (P/F,10\%,3) = -62{,}522 \times 0.7513 = -46{,}969$$

$$PW_strategy3 = -56{,}596 + (-46{,}969) = -Rs.\ 1{,}03{,}565$$

$$AW_3 = -1{,}03{,}565 \times 0.26380 = -Rs.\ 27{,}320$$

Strategy 1 (Defender 1 yr, Challenger 4 yr):

$$PW_def(1yr) = -30{,}000-10{,}000(0.9091)+20{,}000(0.9091) = -30{,}000+10{,}000(0.9091) = -30{,}000+9{,}091 = -20{,}909$$

Challenger from t=1 to t=5 (4 yrs), buy at t=1 for 70,000:

$$PW_chal@t1 = -70{,}000-7{,}000(0.9091)-9{,}000(0.8264)-11{,}500(0.7513)-14{,}500(0.6830)+25{,}000(0.6830)$$

$$= -70{,}000-6{,}364-7{,}438-8{,}640-9{,}904+17{,}075 = -85{,}271$$

$$PW_chal@t0 = -85{,}271 \times 0.9091 = -77{,}520$$

$$PW_strategy1 = -20{,}909-77{,}520 = -Rs.\ 98{,}429 \rightarrow AW_1 = -Rs.\ 25{,}965$$

Summary of all strategies:

Strategy	Description	PW of Total Costs (Rs.)	AW (Rs.)
1	Defend 1yr, Challenge 4yr	-98,429	-25,965 ← Best
3	Defend 3yr, Challenge 2yr	-1,03,565	-27,320
4	Challenge all 5yr	-1,02,346	-26,999

Optimal Strategy: Replace the defender after 1 year (at end of year 1). Use the challenger for years 2 through 5. This gives the minimum AW cost of Rs. 25,965/year over the 5-year study period.

SECTION F: Comprehensive Replacement Problems

Q15. A machine was purchased 5 years ago for Rs. 1,20,000. Its current market value is Rs. 30,000. Expected future O&M cost is Rs. 22,000/year and it will have zero salvage value after 5 more years. A new machine (challenger) costs Rs. 80,000, has O&M of Rs. 10,000/year, a salvage value of Rs. 15,000 after 10 years. Should the defender be replaced? Use AW method with MARR = 12%. [2080 Chaitra Q6, 2078 Baishakh Q6 — 6 Marks]

Answer:

Defender: Original cost Rs. 1,20,000 (sunk cost — IGNORE). Current MV = Rs. 30,000 (first cost for analysis). Annual O&M = Rs. 22,000. Remaining life = 5 years. Salvage = Rs. 0.

Challenger: P = Rs. 80,000. Annual O&M = Rs. 10,000. Life = 10 years. S = Rs. 15,000. MARR = 12%.

AW of Defender (5 years, MARR = 12%):

```
(A/P,12%,5) = 0.27741,  (A/F,12%,5) = 0.15741
AW_D = -30,000(A/P,12%,5) + 0(A/F,12%,5) - 22,000
     = -30,000(0.27741) + 0 - 22,000
     = -8,322 - 22,000 = -Rs. 30,322
```

AW of Challenger (10 years, MARR = 12%):

```
(A/P,12%,10) = 0.17698,  (A/F,12%,10) = 0.05698
AW_C = -80,000(0.17698) + 15,000(0.05698) - 10,000
     = -14,158 + 855 - 10,000 = -Rs. 23,303
```

Comparison:

	Defender	Challenger
AW of Capital Recovery Cost (Rs.)	8,322	13,303
AW of O&M Cost (Rs.)	22,000	10,000
Total AW Cost (Rs.)	30,322	23,303

AW_Challenger (Rs. 23,303/yr) < AW_Defender (Rs. 30,322/yr). The challenger saves Rs. 7,019 per year.

Decision: REPLACE the existing machine with the challenger immediately.

Note: The original purchase price of Rs. 1,20,000 (5 years ago) is completely irrelevant — it is a sunk cost. The analysis correctly uses only Rs. 30,000 (current market value) as the defender's first cost.

Q16. The following data is given for a defender and challenger. Find ESL for both and make replacement decision. MARR = 10%. [2081 Ashwin Q6b — 8 Marks]

Answer:

Year	Defender MV (Rs.)	Defender O&M (Rs.)	Challenger MV (Rs.)	Challenger O&M (Rs.)
0	25,000	—	55,000	—
1	18,000	8,000	42,000	6,000
2	12,000	11,000	31,000	8,000
3	7,500	15,000	22,000	11,000
4	4,000	20,000	15,000	15,000
5	—	—	9,000	20,000

MARR = 10%

(A/P,10%,n): 1→1.10, 2→0.57619, 3→0.40211, 4→0.31547, 5→0.26380)

ESL of Defender (first cost = current MV = Rs. 25,000):

```
TC_D(1)=(25,000-
18,000)(1.10)+18,000(0.10)+8,000=7,700+1,800+8,000=Rs.17,500
TC_D(2): CR=(25,000-12,000)(0.57619)+12,000(0.10)=7,490+1,200=8,690
```

PW_OM=8,000(0.9091)+11,000(0.8264)=7,273+9,090=16,363

AW_OM=16,363×0.57619=9,427; TC_D(2)=8,690+9,427=Rs.18,117

TC_D(3): CR=(25,000-7,500)(0.40211)+7,500(0.10)=7,037+750=7,787

PW_OM=16,363+15,000(0.7513)=16,363+11,270=27,633

AW_OM=27,633×0.40211=11,114; TC_D(3)=7,787+11,114=Rs.18,901

TC_D(4): CR=(25,000-4,000)(0.31547)+4,000(0.10)=6,625+400=7,025

PW_OM=27,633+20,000(0.6830)=27,633+13,660=41,293

AW_OM=41,293×0.31547=13,026; TC_D(4)=7,025+13,026=Rs.20,051

Defender ESL = 1 year, Min TC_Defender = Rs. 17,500

ESL of Challenger (P = Rs. 55,000):

TC_C(1)=(55,000-42,000)(1.10)+42,000(0.10)+6,000=14,300+4,200+6,000=Rs.24,500

TC_C(2): CR=(55,000-31,000)(0.57619)+31,000(0.10)=13,829+3,100=16,929

PW_OM=6,000(0.9091)+8,000(0.8264)=5,455+6,611=12,066

AW_OM=12,066×0.57619=6,952; TC_C(2)=16,929+6,952=Rs.23,881

TC_C(3): CR=(55,000-22,000)(0.40211)+22,000(0.10)=13,270+2,200=15,470

PW_OM=12,066+11,000(0.7513)=12,066+8,264=20,330

AW_OM=20,330×0.40211=8,175; TC_C(3)=15,470+8,175=Rs.23,645

TC_C(4): CR=(55,000-15,000)(0.31547)+15,000(0.10)=12,619+1,500=14,119

PW_OM=20,330+15,000(0.6830)=20,330+10,245=30,575

AW_OM=30,575×0.31547=9,643; TC_C(4)=14,119+9,643=Rs.23,762

TC_C(5): CR=(55,000-9,000)(0.26380)+9,000(0.10)=12,135+900=13,035

PW_OM=30,575+20,000(0.6209)=30,575+12,418=42,993

AW_OM=42,993×0.26380=11,342; TC_C(5)=13,035+11,342=Rs.24,377

Challenger ESL = 3 years, Min TC_Challenger = Rs. 23,645

	Defender	Challenger
ESL	1 year	3 years
Min AW Cost	Rs. 17,500/year	Rs. 23,645/year

Min AW of Defender (Rs. 17,500) < Min AW of Challenger (Rs. 23,645).

Decision: DO NOT REPLACE. Keep the defender for 1 more year (its ESL), then re-evaluate.

Q17. A 4-year old tractor (defender) originally cost Rs. 2,50,000 and has a current resale value of Rs. 70,000. The defender will need O&M costs of Rs. 35,000 in year 1, increasing by Rs. 8,000/year. The defender's market value declines by Rs. 10,000/year. A new tractor (challenger) can be purchased for Rs. 1,80,000 with O&M costs starting at Rs. 15,000, increasing by Rs. 5,000/year. The challenger has a salvage value equal to 15% of the purchase price after each year. MARR = 10%. Find the ESL of each and recommend. [2079 Jestha Q6, 2076 Baisakh Q6 — 8 Marks]

Answer:

Defender: MV_0 = Rs. 70,000 (sunk cost principle — original Rs. 2,50,000 is irrelevant)

Defender MV: decreases Rs. 10,000/yr → MV(1)=60,000; MV(2)=50,000; MV(3)=40,000; MV(4)=30,000

Defender O&M: A1=35,000, G=8,000 → yr1=35,000; yr2=43,000; yr3=51,000; yr4=59,000

Challenger: P = Rs. 1,80,000; O&M: A1=15,000, G=5,000 → yr1=15,000; yr2=20,000; yr3=25,000; yr4=30,000

Challenger salvage = 15% of 1,80,000 = Rs. 27,000/year regardless of year (MV constant at Rs. 27,000)

MARR = 10%

(A/P,10%,n): 1→1.10, 2→0.57619, 3→0.40211, 4→0.31547)

ESL of Defender:

TC_D(1)=(70,000-60,000)(1.10)+60,000(0.10)+35,000=11,000+6,000+35,000=Rs.52,000

TC_D(2): CR=(70,000-50,000)(0.57619)+50,000(0.10)=11,524+5,000=16,524

PW_OM=35,000(0.9091)+43,000(0.8264)=31,819+35,535=67,354

AW_OM=67,354×0.57619=38,812; TC_D(2)=Rs.55,336

TC_D(3): CR=(70,000-40,000)(0.40211)+40,000(0.10)=12,063+4,000=16,063

PW_OM=67,354+51,000(0.7513)=67,354+38,316=1,05,670

AW_OM=1,05,670×0.40211=42,495; TC_D(3)=Rs.58,558

TC_D(4): CR=(70,000-30,000)(0.31547)+30,000(0.10)=12,619+3,000=15,619

PW_OM=1,05,670+59,000(0.6830)=1,05,670+40,297=1,45,967

AW_OM=1,45,967×0.31547=46,063; TC_D(4)=Rs.61,682

Defender ESL = 1 year (TC increases monotonically), Min TC_D = Rs. 52,000

ESL of Challenger (P=1,80,000, MV=27,000 each year):

TC_C(1)=(1,80,000-27,000)(1.10)+27,000(0.10)+15,000=1,68,300+2,700+15,000=Rs.1,86,000

This is very high for n=1. Clearly the challenger needs multiple years. Recalculate for n=3,4,5:

TC_C(3): CR=(1,80,000-27,000)(0.40211)+27,000(0.10)=61,523+2,700=64,223

PW_OM=15,000(0.9091)+20,000(0.8264)+25,000(0.7513)=13,637+16,528+18,783=48,948

AW_OM=48,948×0.40211=19,686; TC_C(3)=64,223+19,686=Rs.83,909

TC_C(4): CR=(1,80,000-27,000)(0.31547)+27,000(0.10)=48,267+2,700=50,967

PW_OM=48,948+30,000(0.6830)=48,948+20,490=69,438

AW_OM=69,438×0.31547=21,904; TC_C(4)=50,967+21,904=Rs.72,871

TC_C(5): CR=(1,80,000-27,000)(0.26380)+27,000(0.10)=40,361+2,700=43,061

PW_OM=69,438+35,000(0.6209)=69,438+21,732=91,170

AW_OM=91,170×0.26380=24,051; TC_C(5)=43,061+24,051=Rs.67,112

TC_C(6): CR=(1,53,000)(0.22961)+27,000(0.10)=35,130+2,700=37,830 [using O&M(6)=40,000]

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TC_C(6) continues to decrease...
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For the purpose of this comparison, using TC_C at n=5 as approximate minimum:

Challenger ESL ≈ 5+ years, Min TC_C ≈ Rs. 67,112

Comparison:

```
Min TC_Defender = Rs. 52,000/year  vs.  Min TC_Challenger = Rs.
                                        67,112/year
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Defender is cheaper by Rs. 15,112/year. Decision: DO NOT REPLACE the tractor. Keep the defender.

The new challenger is significantly more expensive to own annually than the defender at its optimal life, making replacement economically unjustifiable at this time.

SECTION G: Theory and Conceptual Questions

Q18. Differentiate between book value and market value (salvage value) in the context of replacement analysis. Why is market value used? [2078 Chaitra Q5b, 2072 Magh Q5b — 3 Marks]

Answer:

Basis	Book Value	Market Value (Salvage Value)
Definition	Accounting value of an asset on the balance sheet after depreciation	Actual cash price the asset could fetch if sold today in the open market
Determination	Original cost minus accumulated depreciation (accounting method)	Determined by supply and demand in the market
Relevance to replacement	NOT relevant — it is a sunk cost	RELEVANT — represents the opportunity cost of continuing to hold the asset
Example (machine bought for Rs. 5,00,000, 3 yrs old, straight-line dep.)	Book Value = 5,00,000 - 3(60,000) = Rs. 3,20,000	Market Value might be Rs. 1,80,000 (actual resale value)
Used in replacement?	No (sunk cost)	Yes (as defender's first cost)

In replacement analysis, market value is used as the defender's first cost because it represents the true economic cost of keeping the asset. If the defender is kept, the owner foregoes the cash from selling it — this is the opportunity cost. Book value is merely an accounting convention with no cash flow significance.

Q19. Explain the 'opportunity cost approach' and 'cash flow approach' in replacement analysis. Which is recommended? [2077 Chaitra Q5b, 2074 Bhadra Q5a — 3 Marks]

Answer:

Opportunity Cost Approach (Recommended Method)

In the opportunity cost approach, the defender's current market value is treated as its 'investment cost' (first cost for analysis). The rationale is that by choosing to keep the defender, the owner is effectively 'investing' the market value — they are forgoing the cash that could have been obtained by selling the defender. This foregone cash is the opportunity cost of retaining the asset.

```
Defender's first cost = Current Market Value (= Opportunity Cost)
```

This approach correctly reflects the economic reality: keeping the defender is a choice to invest the defender's current value in it, and this investment must earn at least MARR to be justified.

Cash Flow Approach

In the cash flow approach, the defender's market value at the time of replacement decision is listed as a positive cash inflow (proceeds from sale), and the challenger's purchase cost is listed as a negative cash flow. All costs are then expressed in actual cash flow terms.

When properly applied, both approaches give identical results. However, the opportunity cost approach is simpler, more intuitive, and is the standard method taught and used in practice.

Why Opportunity Cost Approach is Recommended:

- It directly aligns with the principle that sunk costs are irrelevant.
- It correctly positions the defender and challenger as equals, with each having an initial 'investment' (defender's MV vs. challenger's purchase price).
- It simplifies computation and avoids confusing cash flow signs.
- It is the standard method in most engineering economics textbooks and professional practice.

Q20. What is the difference between the 'trade-in allowance' and 'market value' of the defender? How should each be treated in replacement analysis? [2076 Baisakh Q5b — 3 Marks]

Answer:

Market Value

Market value (also called fair market value or resale value) is the price at which the defender could be sold to an independent buyer in the open market. It reflects the true economic value of the asset based on supply and demand conditions.

Trade-In Allowance

Trade-in allowance is the amount a seller (manufacturer or dealer) of the challenger will credit the buyer for turning in the defender. It is a marketing/commercial price used to facilitate the sale of the challenger, and it may be inflated above true market value as a sales incentive, or deflated if the seller is not interested in the old equipment.

Treatment in Replacement Analysis

If the trade-in allowance > market value: The difference is effectively a price reduction on the challenger. For replacement analysis, the challenger's 'true' purchase price should be reduced by the excess trade-in allowance, OR use the market value for the defender and the full price for the challenger.

If the trade-in allowance = market value: No adjustment needed. Both give the same result.

Recommendation: Always use the true market value of the defender and the net transaction cost of the challenger. If a trade-in deal is offered, adjust the challenger's cost accordingly.

```
Adjusted challenger first cost = Stated price - (Trade-in allowance -
                                Market value of defender)
```

Example: Defender market value = Rs. 30,000. Dealer offers trade-in of Rs. 40,000 against new machine listed at Rs. 2,00,000. Adjusted challenger cost = 2,00,000 - (40,000-30,000) = Rs. 1,90,000. Use Rs. 1,90,000 as challenger first cost and Rs. 30,000 as defender first cost.

QUICK REFERENCE SUMMARY — Chapter 5 Key Formulas

Formula	Expression	Application
Marginal Cost	$MC_k = (MV_{k-1}-MV_k) + MV_{k-1}\times i + O\&M_k$	Cost of keeping defender one more year
Capital Recovery	$CR(n) = (P-MV_n)(A/P,i,n) + MV_n\times i$	Annual equivalent of capital investment
Total Annual Cost	TC(n) = CR(n) + AW[O&M over n years]	Used to find ESL
ESL	$n^* = \text{argmin}[TC(n)]$	Minimise TC to find optimal life
One-year rule	If $MC_D \leq AW_C(\text{min})$: Keep; else Replace	Quick replacement decision
Capitalized Cost	$CC = P + A/i + C/[(1+i)^{k-1}]$	Permanent structures
Defender first cost	= Current Market Value	NOT original cost or book value

Key Principles:

- Sunk costs (original cost, book value, past depreciation) are ALWAYS irrelevant.
- Defender's first cost = current market value (opportunity cost approach).
- AW method with repeatability assumption is used for unequal-life comparison.
- ESL is the retention period that minimises Total Annual Cost (TC).
- Replace if Min AW(Challenger) < Min AW(Defender); Keep if Min AW(Defender) ≤ Min AW(Challenger).
- Re-evaluate the replacement decision periodically as O&M costs and market values change.

—— End of Chapter 5 Questions & Answers ——

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